A Effects Safety Theorem

The proofs for Effects follow the proofs of System F^{ε} in Effect Handlers, Evidently [?]. Effects is F^{ε} with the expressions of Base, errors, and fixpoints. Its type rules are those of F^{ε}, with the evidence translation left out. This part has four sections: evaluation contexts, substitution, progress, and preservation.

A.1 Conventions

Typing. There is no type variable context. As in F^{ε}, type variables are kept apart by the side conditions $\alpha \notin \mathrm{ftv}(\Gamma)$ of T-E-Perform and $\alpha \notin \mathrm{ftv}(\varepsilon, \tau)$ of T-E-Ops, and every type is assumed well-formed. The rules of Base are lifted to Effects by giving every premise and the conclusion the same effect row ε. The value forms n, b, unit and x are typed by value rules (T-Num, T-Bool, T-Unit, T-E-Var). The effect row is also recorded in effect-bearing syntax: $\lambda^{\varepsilon} x : \tau.e$ records the latent effect of its body, $\mathtt{perform}^{\varepsilon}\ \mathtt{op}\ \tau_s$ records the tail of its operation row, and $\mathtt{handler}_m(h)^{\varepsilon}$ and $\mathtt{handle}_m(h, e)^{\varepsilon}$ record the handler residual row. Thus $\mathtt{handle}_m(h, e)^{\varepsilon}$ checks e at $\langle l \mid \varepsilon \rangle$ and produces effect ε. A tuple of values also has a value rule, T-E-VTuple: from $\Gamma \vdash_{\mathrm{val}} v_1 : \tau_1$ and $\Gamma \vdash_{\mathrm{val}} v_2 : \tau_2$ conclude $\Gamma \vdash_{\mathrm{val}} \langle v_1, v_2 \rangle : \langle \tau_1, \tau_2 \rangle$. Without it, a tuple could not be substituted for a variable in a value position, such as the body of $\Lambda\alpha.v$. In the figure, the rule for λ is named T-E-Abs.

Errors. $\mathtt{error}_p$ is a value, typed by T-E-Error: $\Gamma \vdash_{\mathrm{val}} \mathtt{error}_p : \tau$ for any τ. Step-E-Error aborts the program, $E[\mathtt{error}_p] \Longrightarrow \mathtt{error}_p$ for $E \neq \Box$. We drop the context $\mathtt{error}_E$ of the figure, since the argument of an error is a party, not an expression.

Effect rows and signatures. Rows are lists of labels and may contain duplicates. For a list of labels $L = l_n, ..., l_1$, the row $\langle L \mid \varepsilon \rangle$ is $\langle l_n \mid ... \langle l_1 \mid \varepsilon \rangle ... \rangle$. Unlike F^{ε}, we do not quotient rows by permutation: the rows below are built by the same ordered concatenation. As in [?], operation names are globally unique, so $\mathtt{op} \in \Sigma(l)$ determines l, and T-E-Ops gives a handler for l exactly one clause for every operation in $\Sigma(l)$.

Fixpoints. In Effects, a fixpoint is a recursive value of function or polymorphic type:

T-E-Fix: from $\Gamma, x : \tau \vdash_{\text{val}} v : \tau$ conclude $\Gamma \vdash_{\text{val}} \mu x : \tau.v : \tau$, where τ is $\tau_1 \to^{\varepsilon} \tau_2$ or $\forall \alpha^k.\tau_1$

Step-E-Fix-App: $(\mu x : \tau.v)\ v_0 \rightarrow v[x := \mu x : \tau.v]\ v_0$

Step-E-Fix-TApp: $(\mu x : \tau.v)[\tau_s] \rightarrow v[x := \mu x : \tau.v][\tau_s]$

Remark. The lifted fixpoint of Base, which unfolds $\mu x : \tau.e$ to $e[x := \mu x : \tau.e]$ for any e, does not preserve types once effects are tracked. Let $\tau = \mathsf{unit} \to^{\langle\rangle} \mathsf{num}$ and

$e = \left(\lambda^{\langle l \rangle} y.\lambda^{\langle\rangle} _ : \mathsf{unit}.x\ \mathsf{unit}\right) \mathtt{perform}^{\langle\rangle}\ \mathsf{op}\ \tau_s\ v.$

Then $x : \tau \vdash e : \tau \mid \langle l \rangle$, so $\mu x : \tau.e$ has type τ with effect $\langle l \rangle$. After one step, $\mu x : \tau.e$ sits in the body of $\lambda _ : \mathsf{unit}....$, which has effect $\langle\rangle$. But $\mu x : \tau.e$ can only be typed with the effect of the perform in e, so the result is not well typed. Restricting the body to a value avoids this. A value can be given any effect, and so can a recursive value. The wrap handler of the compiler is such a recursive value.

A.2 Evaluation Contexts

$\lceil E \rceil$ collects the labels of the handlers in E, innermost first. $\mathtt{bop}(E)$ collects the operations they handle.

$$\lceil F_0 \cdot \text{handle } h_1 \cdot F_1 \cdot ... \cdot \text{handle } h_n \cdot F_n \rceil = l_n, ..., l_1 \quad \text{where } h_i : \Sigma(l_i)$$

$$\mathtt{bop}(\square) = \varnothing \qquad \mathtt{bop}(F[E]) = \mathtt{bop}(E) \qquad \mathtt{bop}(\mathtt{handle}_m(h, e)^{\varepsilon}) = \mathtt{bop}(E) \cup \{\mathsf{op} \mid (\mathsf{op} \to f) \in h\}$$

where F is a single frame without a handler. A handler $h : \Sigma(l)$ has a clause for every operation of l (T-E-Ops). So a context E handles $\mathsf{op} \in \Sigma(l)$, i.e. $\mathsf{op} \in \mathtt{bop}(E)$, exactly when l occurs in $\lceil E \rceil$.

The judgment $\Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau_2 \mid \varepsilon$ types a context as a function from the type of its hole to the type of the whole expression, where the whole expression has effect ε.

$$\frac{}{\Gamma \vdash_{\text{ec}} \square : \tau \to \tau \mid \varepsilon}\ \text{CE-Empty}$$

$$\frac{\Gamma \vdash e : \tau_2 \mid \varepsilon \qquad \Gamma \vdash_{\text{ec}} E : \tau_1 \to (\tau_2 \to^{\varepsilon} \tau_3) \mid \varepsilon}{\Gamma \vdash_{\text{ec}} E\, e : \tau_1 \to \tau_3 \mid \varepsilon}\ \text{CE-App1}$$

$$\frac{\Gamma \vdash_{\text{val}} v : \tau_2 \to^{\varepsilon} \tau_3 \qquad \Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau_2 \mid \varepsilon}{\Gamma \vdash_{\text{ec}} v\, E : \tau_1 \to \tau_3 \mid \varepsilon}\ \text{CE-App2}$$

$$\frac{\Gamma \vdash_{\text{ec}} E : \tau_1 \to \forall \alpha.\tau_2 \mid \varepsilon}{\Gamma \vdash_{\text{ec}} E[\tau] : \tau_1 \to \tau_2[\alpha := \tau] \mid \varepsilon}\ \text{CE-TApp}$$

$$\frac{\Gamma \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon \qquad \Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \langle l \mid \varepsilon \rangle}{\Gamma \vdash_{\text{ec}} \texttt{handle}_m(h, e)^{\varepsilon} : \tau_1 \to \tau \mid \varepsilon}\ \text{CE-Handle}$$

$$\frac{\Gamma \vdash_{\text{ec}} E : \tau_1 \to \mathsf{bool} \mid \varepsilon \qquad \Gamma \vdash e_2 : \tau \mid \varepsilon \qquad \Gamma \vdash e_3 : \tau \mid \varepsilon}{\Gamma \vdash_{\text{ec}} \text{if } E \text{ then } e_2 \text{ else } e_3 : \tau_1 \to \tau \mid \varepsilon}\ \text{CE-If}$$

$$\frac{\Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau_a \mid \varepsilon \qquad \Gamma \vdash e : \tau_b \mid \varepsilon}{\Gamma \vdash_{\text{ec}} Eoe : \tau_1 \to \tau_o \mid \varepsilon}\ \text{CE-Op1}$$

$$\frac{\Gamma \vdash v : \tau_a \mid \varepsilon \qquad \Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau_b \mid \varepsilon}{\Gamma \vdash_{\text{ec}} voE : \tau_1 \to \tau_o \mid \varepsilon}\ \text{CE-Op2}$$

Figure 1: Typing of evaluation contexts. The frames $\mathsf{zero?}(E)$, $\langle E, e\rangle$, $\langle v, E\rangle$, $\mathsf{inj.l}(E)$, $\mathsf{inj.r}(E)$ have rules CE-Zero, CE-Tuple1, CE-Tuple2, CE-Inj-Left, CE-Inj-Right of the same form, following the type rule of the frame.

Only CE-Handle changes the effect: the hole of $\texttt{handle}_m(h, e)^{\varepsilon}$ is typed under $\langle l \mid \varepsilon\rangle$.

Following the definitions, for every context

$$\langle \lceil E\, e \rceil \mid \varepsilon\rangle = \langle \lceil E \rceil \mid \varepsilon\rangle \qquad \langle \lceil \texttt{handle}_m(h, e)^{\varepsilon} \rceil \mid \varepsilon\rangle = \langle \lceil E \rceil \mid \langle l \mid \varepsilon\rangle\rangle$$

and likewise for every frame without a handler.

A.3 Statements and proofs

Lemma A.1. (Evaluation context typing) If $\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$ and $\cdot \vdash e : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon\rangle$ then $\cdot \vdash E[e] : \tau \mid \varepsilon$.

Proof. By induction on the derivation of $\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$.

- Case CE-Empty. $E[e] = e$ and $\langle \lceil \square \rceil \mid \varepsilon\rangle = \varepsilon$.

- Case CE-App1, $E = E_0\ e_0$.

  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to (\tau_2 \to^{\varepsilon} \tau) \mid \varepsilon$ CE-App1
  $\cdot \vdash e_0 : \tau_2 \mid \varepsilon$ CE-App1
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \varepsilon \rangle$ given, $\lceil E \rceil = \lceil E_0 \rceil$
  $\cdot \vdash E_0[e] : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ I.H.
  $\cdot \vdash E_0[e]\ e_0 : \tau \mid \varepsilon$ T-E-App

- Case CE-App2, $E = v\ E_0$.

  $\cdot \vdash_{\text{val}} v : \tau_2 \to^{\varepsilon} \tau$ CE-App2
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \tau_2 \mid \varepsilon$ CE-App2
  $\cdot \vdash E_0[e] : \tau_2 \mid \varepsilon$ I.H.
  $\cdot \vdash v : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-Val
  $\cdot \vdash v\ E_0[e] : \tau \mid \varepsilon$ T-E-App

- Case CE-TApp, $E = E_0[\tau_0]$ and $\tau = \tau_2[\alpha := \tau_0]$.

  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \forall \alpha.\tau_2 \mid \varepsilon$ CE-TApp
  $\cdot \vdash E_0[e] : \forall \alpha.\tau_2 \mid \varepsilon$ I.H.
  $\cdot \vdash E_0[e][\tau_0] : \tau_2[\alpha := \tau_0] \mid \varepsilon$ T-E-Tapp

- Case CE-Handle, $E = \texttt{handle}_m(h, e)^{\varepsilon}$.

  $\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ CE-Handle
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \tau \mid \langle l \mid \varepsilon \rangle$ CE-Handle
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \langle l \mid \varepsilon \rangle \rangle$ given, def. of $\lceil \cdot \rceil$
  $\cdot \vdash E_0[e] : \tau \mid \langle l \mid \varepsilon \rangle$ I.H.
  $\cdot \vdash \texttt{handle}_m(h, e)^{\varepsilon} : \tau \mid \varepsilon$ T-E-Handle

- Case CE-If, $E = \text{if } E_0 \text{ then } e_2 \text{ else } e_3$.

  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \mathsf{bool} \mid \varepsilon$ CE-If
  $\cdot \vdash e_2 : \tau \mid \varepsilon, \cdot \vdash e_3 : \tau \mid \varepsilon$ CE-If
  $\cdot \vdash E_0[e] : \mathsf{bool} \mid \varepsilon$ I.H.
  $\cdot \vdash \text{if } E_0[e] \text{ then } e_2 \text{ else } e_3 : \tau \mid \varepsilon$ T-If

- Cases CE-Op1, CE-Op2, CE-Zero, CE-Tuple1, CE-Tuple2, CE-Inj-Left, CE-Inj-Right. As CE-If: the I.H. on E_0, then the lifted rule of Base for the frame.

□

Lemma A.2. (Effect corresponds to the evaluation context) If $\cdot \vdash E[e] : \tau \mid \varepsilon$ then there is τ_1 such that $\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$ and $\cdot \vdash e : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon \rangle$.

Proof. By induction on E.

- Case $E = \square$. Take $\tau_1 = \tau$, and use CE-Empty.
- Case $E = E_0\ e_0$.

  $\cdot \vdash E_0[e] : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ — T-E-App
  $\cdot \vdash e_0 : \tau_2 \mid \varepsilon$ — T-E-App
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to (\tau_2 \to^{\varepsilon} \tau) \mid \varepsilon$ — I.H.
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \varepsilon \rangle$ — I.H.
  $\cdot \vdash_{\text{ec}} E_0\ e_0 : \tau_1 \to \tau \mid \varepsilon$ — CE-App1
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon \rangle$ — $\lceil E \rceil = \lceil E_0 \rceil$
- Case $E = v\ E_0$.

  $\cdot \vdash v : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ — T-E-App
  $\cdot \vdash_{\text{val}} v : \tau_2 \to^{\varepsilon} \tau$ — inversion of T-E-Val
  $\cdot \vdash E_0[e] : \tau_2 \mid \varepsilon$ — T-E-App
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \tau_2 \mid \varepsilon$ — I.H.
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \varepsilon \rangle$ — I.H.
  $\cdot \vdash_{\text{ec}} v\ E_0 : \tau_1 \to \tau \mid \varepsilon$ — CE-App2

  The value v has a function type, so it is not a tuple, and its derivation ends in T-E-Val.
- Case $E = E_0[\tau_0]$.

  $\cdot \vdash E_0[e] : \forall \alpha.\tau_2 \mid \varepsilon$ — T-E-Tapp, $\tau = \tau_2[\alpha := \tau_0]$
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \forall \alpha.\tau_2 \mid \varepsilon$ — I.H.
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \varepsilon \rangle$ — I.H.
  $\cdot \vdash_{\text{ec}} E_0[\tau_0] : \tau_1 \to \tau \mid \varepsilon$ — CE-TApp
- Case $E = \texttt{handle}_m(h, e)^{\varepsilon}$ with $h : \Sigma(l)$.

  $\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ — T-E-Handle
  $\cdot \vdash E_0[e] : \tau \mid \langle l \mid \varepsilon \rangle$ — T-E-Handle
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \tau \mid \langle l \mid \varepsilon \rangle$ — I.H.
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \langle l \mid \varepsilon \rangle \rangle$ — I.H.
  $\cdot \vdash_{\text{ec}} \texttt{handle}_m(h, e)^{\varepsilon} : \tau_1 \to \tau \mid \varepsilon$ — CE-Handle
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon \rangle$ — def. of $\lceil \cdot \rceil$
- Case $E = \text{if } E_0 \text{ then } e_2 \text{ else } e_3$.

  $\cdot \vdash E_0[e] : \mathsf{bool} \mid \varepsilon$ — T-If
  $\cdot \vdash e_2 : \tau \mid \varepsilon, \cdot \vdash e_3 : \tau \mid \varepsilon$ — T-If
  $\cdot \vdash_{\text{ec}} E_0 : \tau_1 \to \mathsf{bool} \mid \varepsilon$ — I.H.
  $\cdot \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil \mid \varepsilon \rangle$ — I.H.

$\cdot \vdash_{\text{ec}} \text{if } E_0 \text{ then } e_2 \text{ else } e_3 : \tau_1 \to \tau \mid \varepsilon$ CE-If

- Cases of the other frames of Base. As if E_0 then e_2 else e_3: invert the lifted rule of the frame, apply the I.H., and close with the context rule of the frame.

□

Lemma A.3. (Well typed operations are handled) If $\cdot \vdash E[\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v] : \tau \mid \langle\rangle$ then $E = E_1[\texttt{handle}_m(h, e)^{\varepsilon}]$ with $\texttt{op} \notin \texttt{bop}(E_2)$ and $(\texttt{op} \to f) \in h$.

Proof. Let $\texttt{op} : \forall\alpha.\tau_1 \to \tau_2 \in \Sigma(l)$.

$\cdot \vdash \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v :$ Lemma A.2

$\tau_3 \mid \langle\lceil E\rceil \mid \langle\rangle\rangle$
$\cdot \vdash \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s :$ T-E-App

$\tau_4 \to^{\langle\lceil E\rceil \mid \langle\rangle\rangle}$

$\tau_3 \mid \langle\lceil E\rceil \mid \langle\rangle\rangle$
$\cdot \vdash_{\text{val}} \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s :$ T-E-Val

$\tau_4 \to^{\langle\lceil E\rceil \mid \langle\rangle\rangle} \tau_3$
$\langle l \mid \varepsilon_0\rangle =$ T-E-Perform, where ε_0 is the explicit annotation of $\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s$

$\langle\lceil E\rceil \mid \langle\rangle\rangle$
$l \in \lceil E\rceil$ the row is not empty

So some handler in E is for l, and it has a clause for op. Take $\texttt{handle}_m(h, e)^{\varepsilon}$ to be the innermost handler of E with a clause for op. Then $(\texttt{op} \to f) \in h$, and no handler in E_2 handles op, so $\texttt{op} \notin \texttt{bop}(E_2)$. □

Lemma A.4. (Effect types are meaningful) If $\cdot \vdash E[\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v] : \tau \mid \varepsilon$ with $\texttt{op} \notin \texttt{bop}(E)$ and $\texttt{op} \in \Sigma(l)$, then $l \in \varepsilon$.

Proof.

$\cdot \vdash \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v : \tau_3 \mid \langle\lceil E\rceil \mid \varepsilon\rangle$ Lemma A.2
$\cdot \vdash_{\text{val}} \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s : \tau_4 \to^{\langle\lceil E\rceil \mid \varepsilon\rangle} \tau_3$ T-E-App, T-E-Val
$\langle l \mid \varepsilon_0\rangle = \langle\lceil E\rceil \mid \varepsilon\rangle$ T-E-Perform
$l \notin \lceil E\rceil$ $\texttt{op} \notin \texttt{bop}(E)$

The row on the left starts with l. If $\lceil E\rceil$ were not empty, the row on the right would start with a label of $\lceil E\rceil$, which is not l. So $\lceil E\rceil$ is empty and $\langle l \mid \varepsilon_0\rangle = \varepsilon$, hence $l \in \varepsilon$. □

Lemma A.5. (Value typing) If $\Gamma \vdash v : \tau \mid \varepsilon$ then $\Gamma \vdash_{\text{val}} v : \tau$. Hence $\Gamma \vdash v : \tau \mid \varepsilon_0$ for every ε_0.

Proof. By induction on v. The second part follows from the first by T-E-Val.

- Case $v = \langle v_1, v_2 \rangle$. The derivation ends in T-E-Val or in the lifted T-Tuple. In the first case we are done. In the second:

  $\Gamma \vdash v_1 : \tau_1 \mid \varepsilon, \Gamma \vdash v_2 : \tau_2 \mid \varepsilon$ — T-Tuple
  $\Gamma \vdash_{\text{val}} v_1 : \tau_1, \Gamma \vdash_{\text{val}} v_2 : \tau_2$ — I.H.
  $\Gamma \vdash_{\text{val}} \langle v_1, v_2 \rangle : \langle \tau_1, \tau_2 \rangle$ — T-E-VTuple

- Any other value. The only rule that types it with an effect is T-E-Val, and its premise is the goal.

□

Lemma A.6. (Weakening) If $\Gamma \vdash e : \tau \mid \varepsilon$ and $x \notin \text{dom}(\Gamma)$ then $\Gamma, x : \tau_s \vdash e : \tau \mid \varepsilon$, and likewise for $\vdash_{\text{val}}$, $\vdash_{\text{ops}}$ and $\vdash_{\text{ec}}$.

Proof. By induction on the derivation. Only T-E-Abs and T-E-Fix extend Γ, with a bound variable that we choose different from x. The side condition $\alpha \notin \text{ftv}(\Gamma, x : \tau_s)$ of T-E-Perform holds by choosing the quantified α of the signature fresh. □

Lemma A.7. (Variable substitution) Let $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v : \tau_1$.
1. If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e : \tau \mid \varepsilon$ then $\Gamma_1, \Gamma_2 \vdash e[x := v] : \tau \mid \varepsilon$.
2. If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} v_0 : \tau$ then $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v_0[x := v] : \tau$.
3. If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ then $\Gamma_1, \Gamma_2 \vdash_{\text{ops}} h[x := v] : \tau \mid l \mid \varepsilon$.

Proof. By mutual induction on the derivations. Let $\Gamma = \Gamma_1, x : \tau_1, \Gamma_2$ and $\Gamma_0 = \Gamma_1, \Gamma_2$. Since v is a value, $v_0[x := v]$ is again a value, so the value rules still apply after substitution.

Part 2.

- Case T-E-Var with $v_0 = x$. Then $\tau = \tau_1$ and $x[x := v] = v$, which has type τ_1 by assumption.
- Case T-E-Var with $v_0 = y \neq x$.

  $y : \tau \in \Gamma_0$ — T-E-Var, $y \neq x$
  $\Gamma_0 \vdash_{\text{val}} y[x := v] : \tau$ — T-E-Var, $y[x := v] = y$

- Case T-E-Abs with $v_0 = \lambda_0^{\varepsilon} y : \tau_2.e$, $y \notin \{x\} \cup \text{fv}(v)$.

  $\Gamma, y : \tau_2 \vdash e : \tau_3 \mid \varepsilon_0$ — T-E-Abs
  $\Gamma_0, y : \tau_2 \vdash_{\text{val}} v : \tau_1$ — Lemma A.6

$\Gamma_0, y : \tau_2 \vdash e[x := v] : \tau_3 \mid \varepsilon_0$ — I.H. 1
$\Gamma_0 \vdash_{\text{val}} \lambda_0^{\varepsilon} y : \tau_2 . e[x := v] : \tau_2 \to^{\varepsilon_0} \tau_3$ — T-E-Abs

- Case T-E-Fix with $v_0 = \mu y : \tau . v_1$. As T-E-Abs, with the I.H. 2 on v_1.
- Case T-E-TAbs with $v_0 = \Lambda \alpha^k . v_1$.

$\Gamma \vdash_{\text{val}} v_1 : \tau_2, k \neq \texttt{lab}$ — T-E-TAbs
$\Gamma_0 \vdash_{\text{val}} v_1[x := v] : \tau_2$ — I.H. 2
$\Gamma_0 \vdash_{\text{val}} \Lambda \alpha^k . v_1[x := v] : \forall \alpha^k . \tau_2$ — T-E-TAbs

- Case T-E-Handler with $v_0 = \texttt{handler}_m(h)^{\varepsilon_0}$.

$\Gamma \vdash_{\text{ops}} h : \tau_2 \mid l \mid \varepsilon_0$ — T-E-Handler
$\Gamma_0 \vdash_{\text{ops}} h[x := v] : \tau_2 \mid l \mid \varepsilon_0$ — I.H. 3
$\Gamma_0 \vdash_{\text{val}} \texttt{handler}_m(h)^{\varepsilon_0} : \left(\mathsf{unit} \to^{\langle l \mid \varepsilon_0 \rangle} \tau_2\right) \to^{\varepsilon_0} \tau_2$ — T-E-Handler

- Case T-E-VTuple with $v_0 = \langle v_1, v_2 \rangle$. The I.H. 2 on both components, then T-E-VTuple.
- Case T-E-Perform with v_0 an effect operation form. It has no **term** variables, so term substitution leaves its type instantiation and explicit tail annotation unchanged. The side condition $\alpha \notin \text{ftv}(\Gamma_0)$ follows from $\alpha \notin \text{ftv}(\Gamma)$.
- Cases T-Num, T-Bool, T-Unit, T-E-Error. No variables.

Part 1.

- Case T-E-Val.

$\Gamma \vdash_{\text{val}} v_0 : \tau$ — T-E-Val
$\Gamma_0 \vdash_{\text{val}} v_0[x := v] : \tau$ — I.H. 2
$\Gamma_0 \vdash v_0[x := v] : \tau \mid \varepsilon$ — T-E-Val

- Case T-E-App with $e = e_1\ e_2$.

$\Gamma \vdash e_1 : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon, \Gamma \vdash e_2 : \tau_2 \mid \varepsilon$ — T-E-App
$\Gamma_0 \vdash e_1[x := v] : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ — I.H. 1
$\Gamma_0 \vdash e_2[x := v] : \tau_2 \mid \varepsilon$ — I.H. 1
$\Gamma_0 \vdash e_1\ e_2[x := v] : \tau \mid \varepsilon$ — T-E-App

- Case T-E-Tapp with $e = e_1[\tau_0]$. The I.H. 1 on e_1, then T-E-Tapp.
- Case T-E-Handle with $e = \texttt{handle}_m(h, e)^{\varepsilon}$. The explicit residual annotation ε is unchanged by term substitution.

$\Gamma \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ — T-E-Handle
$\Gamma \vdash e_1 : \tau \mid \langle l \mid \varepsilon \rangle$ — T-E-Handle
$\Gamma_0 \vdash_{\text{ops}} h[x := v] : \tau \mid l \mid \varepsilon$ — I.H. 3

$\Gamma_0 \vdash e_1[x := v] : \tau \mid \langle l \mid \varepsilon \rangle$ I.H. 1
$\Gamma_0 \vdash (\texttt{handle}_m(h, e)^{\varepsilon})[x := v] : \tau \mid \varepsilon$ T-E-Handle

- Cases of the lifted rules of Base (T-Tuple, T-If, T-Op, T-Zero, T-Inj-Left, T-Inj-Right). The I.H. 1 on every premise, then the same rule.

Part 3.

- Case T-E-Ops. Each operation clause has the type prescribed by T-E-Ops, with the residual row ε on all three arrows. Apply I.H. 2 to every clause body. The result has the same clause type under Γ_0, and T-E-Ops applies because its side conditions mention only ε and τ.

□

Lemma A.8. (Type variable substitution) If $\Gamma \vdash e : \tau \mid \varepsilon$ and $\vdash_{\text{wf}} \tau_0 : k$ then $\Gamma[\alpha^k := \tau_0] \vdash e[\alpha^k := \tau_0] : \tau[\alpha^k := \tau_0] \mid \varepsilon[\alpha^k := \tau_0]$, and likewise for $\vdash_{\text{val}}$ and $\vdash_{\text{ops}}$.

Proof. By induction on the derivation, as Lemma 20 of [?] without the evidence translation.

Bound type variables are renamed away from α and from $\text{ftv}(\tau_0)$, so substitution commutes with them:

$\tau_2[\beta := \tau_s][\alpha := \tau_0] = \tau_2[\alpha := \tau_0]\left[\beta := \tau_{s[\alpha:=\tau_0]}\right]$ when $\beta \notin \{\alpha\} \cup \text{ftv}(\tau_0)$.

The interesting cases are the following.

- Case T-E-Tapp with $e = e_1[\tau_s]$.

$\Gamma \vdash e_1 : \forall\beta.\tau_2 \mid \varepsilon$ T-E-Tapp
$\Gamma[\alpha := \tau_0] \vdash e_1[\alpha := \tau_0] : \forall\beta.\tau_2[\alpha := \tau_0] \mid \varepsilon[\alpha := \tau_0]$ I.H.
$\Gamma[\alpha := \tau_0] \vdash e_1[\alpha := \tau_0]\left[\tau_{s[\alpha:=\tau_0]}\right] : \tau_2[\alpha := \tau_0]\left[\beta := \tau_{s[\alpha:=\tau_0]}\right] \mid \varepsilon[\alpha := \tau_0]$ T-E-Tapp
$\tau_2[\alpha := \tau_0]\left[\beta := \tau_{s[\alpha:=\tau_0]}\right] = \tau_2[\beta := \tau_s][\alpha := \tau_0]$ commuting

- Case T-E-Perform. The type of the operation in $\Sigma(l)$ is closed, so only the instantiation τ_s changes, and the conclusion follows by commuting. The quantified variable of the signature is chosen fresh, so it is still not free in $\Gamma[\alpha := \tau_0]$.
- Case T-E-Ops. The clause variable is chosen fresh for τ_0, so the side condition $\beta \notin \text{ftv}(\varepsilon[\alpha := \tau_0], \tau[\alpha := \tau_0])$ still holds.
- Case T-E-TAbs. The bound variable is renamed, and $k \neq \texttt{lab}$ is unchanged.

All other cases follow from the I.H. and the same rule. □

Lemma A.9. (Canonical forms) Let $\cdot \vdash v : \tau \mid \varepsilon$. Then v is $\texttt{error}_p$, or

- if $\tau = \tau_1 \to^{\varepsilon_0} \tau_2$, v is $\lambda^{\varepsilon_0} x : \tau_1.e$, an operation form $\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s$ whose operation row is ε_0, a handler form $\texttt{handler}_m(h)^{\varepsilon_0}$, or $\mu x : \tau.v_0$;

- if $\tau = \forall\alpha.\tau_s$, v is $\Lambda\alpha.v_0$ or $\mu x : \tau.v_0$;
- if τ is num, bool, unit or $\langle\tau_1, \tau_2\rangle$, v is n, b, unit or $\langle v_1, v_2\rangle$ respectively.

Proof. A variable is not typed in the empty context. By Lemma A.5, $\cdot \vdash_{\text{val}} v : \tau$. Every other value form has exactly one value rule, and the conclusion of that rule fixes the shape of the type. T-E-Error types $\texttt{error}_p$ at any type, and T-E-Fix only gives a function or polymorphic type. □

Lemma A.10. (Steps under contexts) If $e \Longrightarrow e_0$ then $E[e] \Longrightarrow e_0$ for some e_0.

Proof. If the step is Step-E, $e = E_1[e_1]$ with $e_1 \rightarrow e_2$, and $E[e] = (E[E_1])[e_1] \Longrightarrow (E[E_1])[e_2]$ by Step-E. If it is Step-E-Error, $e = E_1\left[\texttt{error}_p\right]$ with $E_1 \neq \square$, so $E[E_1] \neq \square$ and $E[e] \Longrightarrow \texttt{error}_p$ by Step-E-Error. □

Lemma A.11. (Progress with effects) If $\cdot \vdash e_1 : \tau \mid \varepsilon$ then e_1 is a value, or $e_1 \Longrightarrow e_2$, or $e_1 = E[\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v]$ with $\texttt{op} \notin \texttt{bop}(E)$.

Proof. By induction on the typing derivation. Call the last form *stuck on op*.

- Case T-E-Val. e_1 is a value.
- Case T-E-App, $e_1 = e_3\ e_4$.

  $\cdot \vdash e_3 : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App
  $\cdot \vdash e_4 : \tau_1 \mid \varepsilon$ T-E-App

  By the I.H. on e_3:
  - $e_3 \Longrightarrow e_5$. Then e_1 steps by Lemma A.10.
  - $e_3 = E_0[\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v]$. Take $E = E_0\ e_4$. Then $\texttt{bop}(E) = \texttt{bop}(E_0)$, so e_1 is stuck on op.
  - e_3 is a value. By the I.H. on e_4:
    - $e_4 \Longrightarrow e_6$. Then e_1 steps by Lemma A.10.
    - $e_4 = E_0[\texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s\ v]$. Take $E = e_3\ E_0$.
    - e_4 is a value. By Lemma A.9 on e_3:
      - $e_3 = \lambda^{\varepsilon} x : \tau_1.e$. Then $e_1 \Longrightarrow e[x := e_4]$ (S-App).
      - $e_3 = \texttt{handler}_m(h)^{\varepsilon}$. Then $e_1 \Longrightarrow \texttt{handle}_m(h, e)^{\varepsilon}$ (Step-E-Handler).
      - $e_3 = \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_s$. Take $E = \square$; e_1 is stuck on op.
      - $e_3 = \mu x : \tau_f.v_0$. Then $e_1 \Longrightarrow v_0[x := e_3]\ e_4$ (Step-E-Fix-App).
      - $e_3 = \texttt{error}_p$. Error subcase, with the context $\square\ e_4$.
- Case T-E-Tapp, $e_1 = e_3[\tau_1]$ and $\cdot \vdash e_3 : \forall\alpha.\tau_2 \mid \varepsilon$. By the I.H. on e_3:

  - e_3 steps, or is stuck on $\mathtt{op}$ with context E_0. Then e_1 steps, or is stuck on $\mathtt{op}$ with context $E_0[\tau_1]$.
  - e_3 is a value. By Lemma A.9:
    - $e_3 = \Lambda\alpha.v_0$. Then $e_1 \Longrightarrow v_0[\alpha := \tau_1]$ (Step-E-TApp).
    - $e_3 = \mu x : \tau_f.v_0$. Then $e_1 \Longrightarrow v_0[x := e_3][\tau_1]$ (Step-E-Fix-TApp).
    - $e_3 = \mathtt{error}_p$. Error subcase.
- Case T-E-Handle, $e_1 = \mathtt{handle}_m(h, e)^\varepsilon$ with $h : \Sigma(l)$ and $\cdot \vdash e_3 : \tau \mid \langle l \mid \varepsilon \rangle$. By the I.H. on e_3:
  - e_3 steps. Then e_1 steps by Lemma A.10.
  - e_3 is a value. Then $e_1 \Longrightarrow e_3$ (Step-E-Return).
  - $e_3 = E_0[\mathtt{perform}^{\varepsilon_p}\ \mathtt{op}\ \tau_s\ v]$ with $\mathtt{op} \notin \mathtt{bop}(E_0)$.
    - If $(\mathtt{op} \to f) \in h$, then $e_1 \Longrightarrow f[\tau_s]\, v\, k$ with $k = \lambda^\varepsilon x : \tau_2[\alpha := \tau_s].\mathtt{handle}_m(h, e)^\varepsilon$ (Step-E-Perform).
    - Otherwise take $E = \mathtt{handle}_m(h, e)^\varepsilon$. Then $\mathtt{op} \notin \mathtt{bop}(E) = \mathtt{bop}(E_0) \cup \{\mathtt{op}_0 \mid (\mathtt{op}_0 \to f) \in h\}$, so e_1 is stuck on $\mathtt{op}$.
- Case T-If, $e_1 = \text{if } e_3 \text{ then } e_4 \text{ else } e_5$. If e_3 steps or is stuck, so is e_1, with the context if E_0 then e_4 else e_5. If e_3 is a value of type bool, it is true or false and S-If-True or S-If-False applies, or it is $\mathtt{error}_p$ (error subcase).
- Cases T-Op and T-Zero. As T-E-App: evaluate the operands from left to right; if all are values they are numbers or booleans by Lemma A.9 and S-Op or S-Zero applies, or one of them is $\mathtt{error}_p$ (error subcase).
- Cases T-Tuple, T-Inj-Left and T-Inj-Right. If the operands are values, a tuple is a value; $\mathsf{inj.l}(\langle v_1, v_2 \rangle)$ and $\mathsf{inj.r}(\langle v_1, v_2 \rangle)$ step by S-Inj-Left and S-Inj-Right; an operand $\mathtt{error}_p$ is the error subcase.

□

Theorem A.12. (Effects Progress) If $\cdot \vdash e_1 : \tau \mid \langle\rangle$ then e_1 is a value or $e_1 \Longrightarrow e_2$.

Proof. By Lemma A.11, e_1 is a value, or steps, or $e_1 = E[\mathtt{perform}^{\varepsilon_p}\ \mathtt{op}\ \tau_s\ v]$ with $\mathtt{op} \notin \mathtt{bop}(E)$ and $\mathtt{op} \in \Sigma(l)$. The last case is impossible:

$\cdot \vdash E[\mathtt{perform}^{\varepsilon_p}\ \mathtt{op}\ \tau_s\ v] : \tau \mid \langle\rangle$	Given
$l \in \langle\rangle$	Lemma A.4
Contradiction	$\langle\rangle$ is empty

□

Lemma A.13. (Redex preservation) If $\cdot \vdash e_1 : \tau \mid \varepsilon$ and $e_1 \rightarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \varepsilon$.

Proof. By cases on $e_1 \rightarrow e_2$.

- Case S-App, $(\lambda^\varepsilon x : \tau_1.e)\ v \rightarrow e[x := v]$.

$\cdot \vdash \lambda^{\varepsilon} x : \tau_1 . e : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App

The λ annotation is the body effect ε. T-E-Abs

$\cdot \vdash v : \tau_1 \mid \varepsilon$ T-E-App

$x : \tau_1 \vdash e : \tau \mid \varepsilon$ T-E-Val, T-E-Abs

$\cdot \vdash_{\text{val}} v : \tau_1$ Lemma A.5

$\cdot \vdash e[x := v] : \tau \mid \varepsilon$ Lemma A.7

- Case Step-E-TApp, $(\Lambda \alpha^k . v)[\tau_0] \rightarrow v[\alpha := \tau_0]$, with $\tau = \tau_1[\alpha := \tau_0]$.

  $\cdot \vdash \Lambda \alpha^k . v : \forall \alpha^k . \tau_1 \mid \varepsilon, \vdash_{\text{wf}} \tau_0 : k$ T-E-Tapp

  $\cdot \vdash_{\text{val}} v : \tau_1$ T-E-Val, T-E-TAbs

  $\cdot \vdash_{\text{val}} v[\alpha := \tau_0] : \tau_1[\alpha := \tau_0]$ Lemma A.8

  $\cdot \vdash v[\alpha := \tau_0] : \tau \mid \varepsilon$ T-E-Val

  The context is empty, and α is bound in $\Lambda \alpha . v$, so $\alpha \notin \text{ftv}(\varepsilon)$ and $\varepsilon[\alpha := \tau_0] = \varepsilon$.

- Case Step-E-Fix-App, $(\mu x : \tau_f . v)\ v_0 \rightarrow v[x := \mu x : \tau_f . v]\ v_0$, with $\tau_f = \tau_1 \to^{\varepsilon} \tau$.

  $\cdot \vdash_{\text{val}} \mu x : \tau_f . v : \tau_f$ T-E-App, T-E-Val

  $\cdot \vdash v_0 : \tau_1 \mid \varepsilon$ T-E-App

  $x : \tau_f \vdash_{\text{val}} v : \tau_f$ T-E-Fix

  $\cdot \vdash_{\text{val}} v[x := \mu x : \tau_f . v] : \tau_f$ Lemma A.7

  $\cdot \vdash v[x := \mu x : \tau_f . v]\ v_0 : \tau \mid \varepsilon$ T-E-Val, T-E-App

- Case Step-E-Fix-TApp. As Step-E-Fix-App, closing with T-E-Tapp.

- Case Step-E-Handler, $\texttt{handler}_m(h)^{\varepsilon}\ v \rightarrow \texttt{handle}_m(h, e)^{\varepsilon}$.

  $\cdot \vdash \texttt{handler}_m(h)^{\varepsilon} : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App

  $\cdot \vdash v : \tau_1 \mid \varepsilon$ T-E-App

  $\cdot \vdash_{\text{val}} \texttt{handler}_m(h)^{\varepsilon} : (\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau) \to^{\varepsilon} \tau$ T-E-Val

  $\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon, \tau_1 = \mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau$ T-E-Handler

  $\cdot \vdash v : \mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau \mid \langle l \mid \varepsilon \rangle$ Lemma A.5

  $\cdot \vdash v\ \mathsf{unit} : \tau \mid \langle l \mid \varepsilon \rangle$ T-Unit, T-E-App

  $\cdot \vdash \texttt{handle}_m(h, e)^{\varepsilon} : \tau \mid \varepsilon$ T-E-Handle

- Case Step-E-Return, $\texttt{handle}_m(h, e)^{\varepsilon} \rightarrow v$.

  $\cdot \vdash v : \tau \mid \langle l \mid \varepsilon \rangle$ T-E-Handle

  $\cdot \vdash v : \tau \mid \varepsilon$ Lemma A.5

- Case Step-E-Perform, $\texttt{handle}_m(h, e)^{\varepsilon} \rightarrow f[\tau_0]\ v\ k$, where $\texttt{op} \notin \texttt{bop}(E)$, $(\texttt{op} \to f) \in h$, $\texttt{op} : \forall \alpha . \tau_1 \to \tau_2 \in \Sigma(l)$ and $k = \lambda^{\varepsilon} x : \tau_2[\alpha := \tau_0] . \texttt{handle}_m(h, e)^{\varepsilon}$.

  $\cdot \vdash_{\text{ec}} \texttt{handle}_m(h, e)^{\varepsilon} : \tau_3 \to \tau \mid \varepsilon$ Lemma A.2

$\cdot \vdash \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_0\ v : \tau_3 \mid \langle \lceil E \rceil \mid \langle l \mid \varepsilon \rangle \rangle$ — Lemma A.2
$\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon, \cdot \vdash_{\text{ec}} E : \tau_3 \to \tau \mid \langle l \mid \varepsilon \rangle$ — CE-Handle
$\cdot \vdash_{\text{val}} f : \forall \alpha . \tau_1 \to^{\varepsilon} ((\tau_2 \to^{\varepsilon} \tau) \to^{\varepsilon} \tau)$ — T-E-Ops
$\alpha \notin \text{ftv}(\varepsilon, \tau)$ — T-E-Ops
$\cdot \vdash f[\tau_0] : \tau_1[\alpha := \tau_0] \to^{\varepsilon} ((\tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau) \to^{\varepsilon} \tau) \mid \varepsilon$ — T-E-Val, T-E-Tapp
$\cdot \vdash_{\text{val}} \texttt{perform}^{\varepsilon_p}\ \texttt{op}\ \tau_0 : \tau_4 \to^{\langle \lceil E \rceil \mid \langle l \mid \varepsilon \rangle \rangle} \tau_3$ — T-E-App, T-E-Val
$\tau_4 = \tau_1[\alpha := \tau_0], \tau_3 = \tau_2[\alpha := \tau_0]$ — T-E-Perform
$\cdot \vdash v : \tau_1[\alpha := \tau_0] \mid \varepsilon$ — T-E-App, Lemma A.5
$\cdot \vdash f[\tau_0]\ v : (\tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau) \to^{\varepsilon} \tau \mid \varepsilon$ — T-E-App
$x : \tau_2[\alpha := \tau_0] \vdash x : \tau_2[\alpha := \tau_0] \mid \langle \lceil E \rceil \mid \langle l \mid \varepsilon \rangle \rangle$ — T-E-Var, T-E-Val
$x : \tau_2[\alpha := \tau_0] \vdash_{\text{ec}} \texttt{handle}_m(h, e)^{\varepsilon} : \tau_2[\alpha := \tau_0] \to \tau \mid \varepsilon$ — Lemma A.6
$x : \tau_2[\alpha := \tau_0] \vdash \texttt{handle}_m(h, e)^{\varepsilon} : \tau \mid \varepsilon$ — Lemma A.1
$\cdot \vdash k : \tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau \mid \varepsilon$ — T-E-Abs, T-E-Val
$\cdot \vdash f[\tau_0]\ v\ k : \tau \mid \varepsilon$ — T-E-App

Substituting for α leaves ε and τ unchanged, because $\alpha \notin \text{ftv}(\varepsilon, \tau)$.

- Cases S-If-True, S-If-False, S-Inj-Left, S-Inj-Right, S-Op, S-Zero. As in [?], with every judgment at effect ε.

□

Theorem A.14. (Effects Preservation) If $\cdot \vdash e_1 : \tau \mid \varepsilon$ and $e_1 \Longrightarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \varepsilon$.

Proof. By cases on the step.

- Case Step-E, $e_1 = E[e_a]$, $e_2 = E[e_b]$ and $e_a \rightarrow e_b$.

  $\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$ — Lemma A.2
  $\cdot \vdash e_a : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon \rangle$ — Lemma A.2
  $\cdot \vdash e_b : \tau_1 \mid \langle \lceil E \rceil \mid \varepsilon \rangle$ — Lemma A.13
  $\cdot \vdash E[e_b] : \tau \mid \varepsilon$ — Lemma A.1

- Case Step-E-Error, $E[\texttt{error}_p] \Longrightarrow \texttt{error}_p$.

  $\cdot \vdash_{\text{val}} \texttt{error}_p : \tau$ — T-E-Error
  $\cdot \vdash \texttt{error}_p : \tau \mid \varepsilon$ — T-E-Val

□

Theorem A.15. (Effects Type Safety) If $\cdot \vdash e : \tau \mid \langle\rangle$ and $e \Longrightarrow^* e_0$ then e_0 is a value or $e_0 \Longrightarrow e_0$ for some e_0.

Proof. By Theorem A.14 on each step, $\cdot \vdash e_0 : \tau \mid \langle\rangle$. Then by Theorem A.12. □